Chapter 4

Cubes and Cube-Roots

Class - 8 Concise Mathematics Selina

Exercise 4(A)

1. 1.25
2. 0.125
3. 12.5
4. 0.0125

Answer

$(0.5)^3$

$= 0.5 \times 0.5 \times 0.5$

$= 0.125$

Hence, option 2 is the correct option.

Question 1(ii)

The cube of -4 is:

1. 16
2. 12
3. -64
4. 64

Answer

$(-4)^3$

$= (-4) \times (-4) \times (-4)$

$= -64$

Hence, option 3 is the correct option.

Question 1(iii)

The smallest number by which 72 must be multiplied to obtain a perfect cube is:

1. 3
2. 2
3. 4
4. 6

Answer

Finding prime factors of 72

$$\begin{array}{r|l} 2 & 72 \\ \hline 2 & 36 \\ \hline 2 & 18 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

72 = (2 x 2 x 2) x 3 x 3

Since the prime factor 3 is not in pair.

The given number should be multiplied by 3.

Hence, option 1 is the correct option.

Question 1(iv)

The smallest number by which 81 be divided to obtain a perfect cube is :

1. 1
2. 3

3. 9

4. none of the above

Answer

Finding prime factors of 81

3	81
3	27
3	9
	3

81 = (3 x 3 x 3) x 3

Since the prime factor 3 is not in pair.

The given number should be divided by 3.

Hence, option 2 is the correct option.

Question 2(i)

Find the cube of 7.

Answer

$(7)^3$

$= 7 \times 7 \times 7$

$= 343$

$(7)^3 = 343$

Question 2(ii)

Find the cube of 11.

Answer

$(11)^3$

$= 11 \times 11 \times 11$

$= 1331$

$(11)^3 = 1331$

Question 2(iii)

Find the cube of 16.

Answer

$(16)^3$

$= 16 \times 16 \times 16$

$= 4096$

$(16)^3 = 4096$

Question 2(iv)

Find the cube of 23.

Answer

$(23)^3$

$= 23 \times 23 \times 23$

$= 12167$

$(23)^3 = 12167$

Question 2(v)

Find the cube of 31.

Answer

$(31)^3$

$= 31 \times 31 \times 31$

$= 29791$

$(31)^3 = 29791$

Question 2(vi)

Find the cube of 42.

Answer

$(42)^3$

$= 42 \times 42 \times 42$

$= 74088$

$(42)^3 = 74088$

Question 2(vii)

Find the cube of 54.

Answer

$(54)^3$

$= 54 \times 54 \times 54$

$= 157464$

$(54)^3 = 157464$

Question 3

Find which of the following are perfect cubes?

(i) 243

(ii) 588

(iii) 1331

(iv) 24000

(v) 1728

(vi) 1938

Answer

(i) 243

Finding prime factors of 243

3	243
3	81
3	27
3	9
	3

243 = (3 x 3 x 3) x 3 x 3

Since the prime factor 3 is not in triplets,

Hence, 243 is not a perfect cube.

(ii) 588

Finding prime factors of 588

2	588
2	294
3	147
7	49
	7

588 = 2 x 2 x 3 x 7 x 7

Since the prime factor 2, 3 and 7 are not in triplets,

Hence, 588 is not a perfect cube.

(iii) 1331

Finding prime factors of 1331

$$\begin{array}{r|r} 11 & 1331 \\ \hline 11 & 121 \\ \hline & 11 \end{array}$$

1331 = (11 x 11 x 11)

Since the prime factor 11 forms a triplet,

Hence, 1331 is a perfect cube.

(iv) 24000

Finding prime factors of 24000

$$\begin{array}{r|r} 2 & 24000 \\ \hline 2 & 12000 \\ \hline 2 & 6000 \\ \hline 2 & 3000 \\ \hline 3 & 1500 \\ \hline 2 & 500 \\ \hline 2 & 250 \\ \hline 5 & 125 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

24000 = (2 x 2 x 2) x (2 x 2 x 2) x 3 x (5 x 5 x 5)

Since the prime factor 3 is not in triplets,

Hence, 24000 is not a perfect cube.

(v) 1728

Finding prime factors of 1728

2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
	3

1728 = (2 x 2 x 2) x (2 x 2 x 2) x (3 x 3 x 3)

Since the prime factor 2 and 3 are in triplets,

Hence, 1728 is a perfect cube.

(vi) 1938

Finding prime factors of 1938

2	1938
3	969
17	323
	19

1938 = 2 x 3 x 17 x 19

Since the prime factor 2, 3, 17 and 19 are not in triplets,

Hence, 1938 is not a perfect cube.

Question 4(i)

Find the cube of 2.1.

Answer

$(2.1)^3$

$= 2.1 \times 2.1 \times 2.1$

$= 9.261$

$(2.1)^3 = 9.261$

Question 4(ii)

Find the cube of 0.4.

Answer

$(0.4)^3$

$= 0.4 \times 0.4 \times 0.4$

$= 0.064$

$(0.4)^3 = 0.064$

Question 4(iii)

Find the cube of 1.6.

Answer

$(1.6)^3$

$= 1.6 \times 1.6 \times 1.6$

$= 4.096$

$(1.6)^3 = 4.096$

Question 4(iv)

Find the cube of 2.5.

Answer

$(2.5)^3$

$= 2.5 \times 2.5 \times 2.5$

$= 15.625$

$(2.5)^3 = 15.625$

Question 4(v)

Find the cube of 0.12.

Answer

$(0.12)^3$

$= 0.12 \times 0.12 \times 0.12$

$= 0.001728$

$(0.12)^3 = 0.001728$

Question 4(vi)

Find the cube of 0.02.

Answer

$(0.02)^3$

$= 0.02 \times 0.02 \times 0.02$

$= 0.000008$

$(0.02)^3 = 0.000008$

Question 4(vii)

Find the cube of 0.8.

Answer

$(0.8)^3$

$= 0.8 \times 0.8 \times 0.8$

$= 0.512$

$(0.8)^3 = 0.512$

Question 5(i)

Find the cube of $\frac{3}{7}$

Answer

$$\left(\frac{3}{7}\right)^3$$

$$= \left(\frac{3}{7} \times \frac{3}{7} \times \frac{3}{7}\right)$$

$$= \frac{27}{343}$$

$$\left(\frac{3}{7}\right)^3 = \frac{27}{343}$$

Question 5(ii)

Find the cube of $\frac{8}{9}$

Answer

$$\left(\frac{8}{9}\right)^3$$

$$= \left(\frac{8}{9} \times \frac{8}{9} \times \frac{8}{9}\right)$$

$$= \frac{512}{729}$$

$$\left(\frac{8}{9}\right)^3 = \frac{512}{729}$$

Question 5(iii)

Find the cube of $\frac{10}{13}$

Answer

$$\left(\frac{10}{13}\right)^3$$

$$= \left(\frac{10}{13} \times \frac{10}{13} \times \frac{10}{13}\right)$$

$$= \frac{1000}{2197}$$

$$\left(\frac{10}{13}\right)^3 = \frac{1000}{2197}$$

Question 5(iv)

Find the cube of $1\frac{2}{7}$

Answer

$$\left(1\frac{2}{7}\right)^3$$

$$= \left(\frac{9}{7}\right)^3$$

$$= \left(\frac{9}{7} \times \frac{9}{7} \times \frac{9}{7}\right)$$

$$= \frac{729}{343}$$

$$= 2\frac{43}{343}$$

$$\left(1\frac{2}{7}\right)^3 = 2\frac{43}{343}$$

Question 5(v)

Find the cube of $2\frac{1}{2}$

Answer

$$\left(2\frac{1}{2}\right)^3$$

$$= \left(\frac{5}{2}\right)^3$$

$$= \left(\frac{5}{2} \times \frac{5}{2} \times \frac{5}{2}\right)$$

$$= \frac{125}{8}$$

$$= 15\frac{5}{8}$$

$$\left(2\frac{1}{2}\right)^3 = 15\frac{5}{8}$$

Question 6(i)

Find the cube of -3.

Answer

$$(-3)^3$$

$$= (-3) \times (-3) \times (-3)$$

$$= (-27)$$

$$(-3)^3 = -27$$

Question 6(ii)

Find the cube of -7.

Answer

$$(-7)^3$$

$$= (-7) \times (-7) \times (-7)$$

$$= (-343)$$

$$(-7)^3 = -343$$

Question 6(iii)

Find the cube of -12.

Answer

$$(-12)^3$$

$$= (-12) \times (-12) \times (-12)$$

$$= (-1728)$$

$$(-12)^3 = -1728$$

Question 6(iv)

Find the cube of -18

Answer

$(-18)^3$

$= (-18) \times (-18) \times (-18)$

$= (-5832)$

$(-18)^3 = -5832$

Question 6(v)

Find the cube of -25

Answer

$(-25)^3$

$= (-25) \times (-25) \times (-25)$

$= (-15625)$

$(-25)^3 = -15625$

Question 6(vi)

Find the cube of -30

Answer

$(-30)^3$

$= (-30) \times (-30) \times (-30)$

$= (-27000)$

$(-30)^3 = -27000$

Question 6(vii)

Find the cube of -50

Answer

$(-50)^3$

$= (-50) \times (-50) \times (-50)$

$= (-125000)$

$(-50)^3 = -125000$

Question 7

Which of the following are cubes of:

(i) an even number

(ii) an odd number.

216, 729, 3375, 8000, 125, 343, 4096 and 9261.

Answer

(i) 216, 8000 and 4096 are the cubes of an even number

Reason — Cubes of even natural numbers are even.

(ii) 729, 3375, 125, 343 and 9261 are the cubes of an odd number.

Reason — Cubes of odd natural numbers are odd.

Question 8

Find the least number by which 1323 must be multiplied so that the product is a perfect cube.

Answer

Finding prime factors of 1323

3	1323
3	441
3	147
7	49
	7

$$1323 = (3 \times 3 \times 3) \times 7 \times 7$$

Since the prime factor 7 does not form a triplet,

Hence, 1323 should be multiplied with 7 so that the product is a perfect cube.

Question 9

Find the smallest number by which 8768 must be divided so that the quotient is a perfect cube.

Answer

Finding prime factors of 8768

2	8768
2	4384
2	2192
2	1096
2	548
2	274
	137

$$8768 = (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times 137$$

Since the prime factor 137 is not in triplet,

8768 will be divided by 137 so that the quotient is a perfect cube.

Question 10

Find the smallest number by which 27783 should be multiplied to get a perfect cube number.

Answer

Finding prime factors of 27783

3	27783
3	9261
3	3087
3	1029
7	343
7	49
	7

$$27783 = (3 \times 3 \times 3) \times 3 \times (7 \times 7 \times 7)$$

Since the prime factor 3 is not in triplets,

3 x 3 = 9 should be multiplied with 27783 so that the product is a perfect cube.

Question 11

With what least number should 8640 be divided so that the quotient is a perfect cube?

Answer

Finding prime factors of 8640

2	8640
2	4320
2	2160
2	1080
2	540
2	270
3	135
3	45
3	15
	5

$$8640 = (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (3 \times 3 \times 3) \times 5$$

Since the prime factor 5 is not in triplets,

8640 will be divided by 5 so that the product is a perfect cube.

Question 12

Which is the smallest number that should be multiplied to 77175 to make it a perfect cube?

Answer

Finding prime factors of 77175

3	77175
3	25725
5	8575
5	1715
7	343
7	49
	7

$$77175 = 3 \times 3 \times 5 \times 5 \times (7 \times 7 \times 7)$$

Since the prime factors 3 and 5 are not in triplets,

3 x 5 = 15 should be multiplied with 77175 so that the product is a perfect cube.

Exercise 4(B)

Question 1(i)

The cube root of 0.000027 is :

1. 0.03

2. 0.003

3. 0.3

4. 0.00003

Answer

$\sqrt[3]{0.000027}$

$= \sqrt[3]{\dfrac{27}{1000000}}$

$= \dfrac{\sqrt[3]{27}}{\sqrt[3]{1000000}}$

$= \dfrac{\sqrt[3]{3 \times 3 \times 3}}{\sqrt[3]{100 \times 100 \times 100}}$

$= \dfrac{3}{100}$

$= 0.03$

Hence, option 1 is the correct option.

Question 1(ii)

The cube root of -0.064 is:

1. -0.8

2. 0.8

3. 0.4

4. -0.4

Answer

$$\sqrt[3]{-0.064}$$

$$= \sqrt[3]{-\frac{64}{1000}}$$

$$= -\frac{\sqrt[3]{64}}{\sqrt[3]{1000}}$$

$$= -\frac{\sqrt[3]{4 \times 4 \times 4}}{\sqrt[3]{10 \times 10 \times 10}}$$

$$= -\frac{4}{10}$$

$$= -0.4$$

Hence, option 4 is the correct option.

Question 2(i)

Find the cube-root of 64.

Answer

Finding prime factors of 64:

2	64
2	32
2	16
2	8
2	4
	2

$\sqrt[3]{64}$

$= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)}$

$= 2 \times 2$

$= 4$

Hence, $\sqrt[3]{64} = 4$

Question 2(ii)

Find the cube-root of 343.

Answer

Finding prime factors of 343:

7	343
7	49
	7

$$\sqrt[3]{343}$$

$$= \sqrt[3]{7 \times 7 \times 7}$$

$$= 7$$

Hence, $\sqrt[3]{343} = 7$

Question 2(iii)

Find the cube-root of 729.

Answer

Finding prime factors of 729:

3	729
3	243
3	81
3	27
3	9
	3

$$\sqrt[3]{729}$$

$$= \sqrt[3]{(3 \times 3 \times 3) \times (3 \times 3 \times 3)}$$

$$= 3 \times 3$$

$$= 9$$

Hence, $\sqrt[3]{729} = 9$

Question 2(iv)

Find the cube-root of 1728.

Answer

Finding prime factors of 1728:

2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
	3

$\sqrt[3]{1728}$

$= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (3 \times 3 \times 3)}$

$= 2 \times 2 \times 3$

$= 12$

Hence, $\sqrt[3]{1728} = 12$

Question 2(v)

Find the cube-root of 9261.

Answer

Finding prime factors of 9261:

3	9261
3	3087
3	1029
7	343
7	49
	7

$\sqrt[3]{9261}$

$= \sqrt[3]{(3 \times 3 \times 3) \times (7 \times 7 \times 7)}$

$= 3 \times 7$

$= 21$

Hence, $\sqrt[3]{9261} = 21$

Question 2(vi)

Find the cube-roots of 4096.

Answer

Finding prime factors of 4096:

2	4096
2	2048
2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

$$\sqrt[3]{4096}$$

$$= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)}$$

$$= 2 \times 2 \times 2 \times 2$$

$$= 16$$

Hence, $\sqrt[3]{4096} = 16$

Question 2(vii)

Find the cube-roots of 8000.

Answer

Finding prime factors of 8000:

2	8000
2	4000
2	2000
2	1000
2	500
2	250
5	125
5	25
	5

$\sqrt[3]{8000}$

$= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (5 \times 5 \times 5)}$

$= 2 \times 2 \times 5$

$= 20$

Hence, $\sqrt[3]{8000} = 20$

Question 2(viii)

Find the cube-roots of 3375.

Answer

Finding prime factors of 3375:

$$\begin{array}{r|l} 3 & 3375 \\ \hline 3 & 1125 \\ \hline 3 & 375 \\ \hline 5 & 125 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

$\sqrt[3]{3375}$

$= \sqrt[3]{(3 \times 3 \times 3) \times (5 \times 5 \times 5)}$

$= 3 \times 5$

$= 15$

Hence, $\sqrt[3]{3375} = 15$

Question 3(i)

Find the cube-roots of $\dfrac{27}{64}$

Answer

Prime factors of 27:

$$\begin{array}{r|l} 3 & 27 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

Prime factors of 64:

2	64
2	32
2	16
2	8
2	4
	2

$$\sqrt[3]{\dfrac{27}{64}}$$

$$= \dfrac{\sqrt[3]{27}}{\sqrt[3]{64}}$$

$$= \dfrac{\sqrt[3]{3 \times 3 \times 3}}{\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)}}$$

$$= \dfrac{3}{2 \times 2}$$

$$= \dfrac{3}{4}$$

Hence, $\sqrt[3]{\dfrac{27}{64}} = \dfrac{3}{4}$

Question 3(ii)

Find the cube-roots of $\dfrac{125}{216}$

Answer

Prime factors of 125:

$$\begin{array}{r|r} 5 & 125 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

Prime factors of 216:

$$\begin{array}{r|r} 2 & 216 \\ \hline 2 & 108 \\ \hline 2 & 54 \\ \hline 3 & 27 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

$$\sqrt[3]{\frac{125}{216}}$$

$$= \frac{\sqrt[3]{125}}{\sqrt[3]{216}}$$

$$= \frac{\sqrt[3]{5 \times 5 \times 5}}{\sqrt[3]{(2 \times 2 \times 2) \times (3 \times 3 \times 3)}}$$

$$= \frac{5}{2 \times 3}$$

$$= \frac{5}{6}$$

Hence, $\sqrt[3]{\frac{125}{216}} = \frac{5}{6}$

Question 3(iii)

Find the cube-roots of $\dfrac{343}{512}$

Answer

Prime factors of 343:

7	343
7	49
	7

Prime factors of 512:

2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

$$\sqrt[3]{\dfrac{343}{512}}$$

$$= \dfrac{\sqrt[3]{343}}{\sqrt[3]{512}}$$

$$= \frac{\sqrt[3]{7 \times 7 \times 7}}{\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)}}$$

$$= \frac{7}{2 \times 2 \times 2}$$

$$= \frac{7}{8}$$

Hence, $\sqrt[3]{\frac{343}{512}} = \frac{7}{8}$

Question 3(iv)

Find the cube-roots of 64 x 729

Answer

Prime factors of 64:

3	27
3	9
	3

Prime factors of 729:

3	729
3	243
3	81
3	27
3	9
	3

$$\sqrt[3]{64 \times 729}$$

$$= \sqrt[3]{64} \times \sqrt[3]{729}$$

$$= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)} \times \sqrt[3]{(3 \times 3 \times 3) \times (3 \times 3 \times 3)}$$

$$= (2 \times 2) \times (3 \times 3)$$

$$= (4) \times (9)$$

$$= 36$$

Hence, $\sqrt[3]{64 \times 729} = 36$

Question 3(v)

Find the cube-roots of 64 x 27

Answer

Prime factors of 64:

$$\begin{array}{r|l} 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline & 2 \end{array}$$

Prime factors of 27:

$$\begin{array}{r|l} 3 & 27 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

$\sqrt[3]{64 \times 27}$

$= \sqrt[3]{64} \times \sqrt[3]{27}$

$= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)} \times \sqrt[3]{(3 \times 3 \times 3)}$

$= (2 \times 2) \times (3)$

$= 4 \times 3$

$= 12$

Hence, $\sqrt[3]{64 \times 27} = 12$

Question 3(vi)

Find the cube-roots of 729 x 8000

Answer

Prime factors of 729:

$$\begin{array}{r|l} 3 & 729 \\ \hline 3 & 243 \\ \hline 3 & 81 \\ \hline 3 & 27 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

Prime factors of 8000:

$$\begin{array}{r|r} 2 & 8000 \\ \hline 2 & 4000 \\ \hline 2 & 2000 \\ \hline 2 & 1000 \\ \hline 2 & 500 \\ \hline 2 & 250 \\ \hline 5 & 125 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

$$\sqrt[3]{729 \times 8000}$$

$$= \sqrt[3]{729} \times \sqrt[3]{8000}$$

$$= \sqrt[3]{(3 \times 3 \times 3) \times (3 \times 3 \times 3)} \times$$
$$\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (5 \times 5 \times 5)}$$

$= (3 \times 3) \times (2 \times 2 \times 5)$

$= 9 \times 20$

$= 180$

Hence, $\sqrt[3]{729 \times 8000} = 180$

Question 3(vii)

Find the cube-roots of 3375 x 512

Answer

Prime factors of 3375:

3	3375
3	1125
3	375
5	125
5	25
	5

Prime factors of 512:

2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

$\sqrt[3]{3375 \times 512}$

$= \sqrt[3]{3375} \times \sqrt[3]{512}$

$= \sqrt[3]{(3 \times 3 \times 3) \times (5 \times 5 \times 5)} \times \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)}$

$= (3 \times 5) \times (2 \times 2 \times 2)$

$= (15) \times (8)$

$= 120$

Hence, $\sqrt[3]{3375 \times 512} = 120$

Question 4(i)

Find the cube-roots of -216.

Answer

Prime factors of 216:

2	216
2	108
2	54
3	27
3	9
	3

$$\sqrt[3]{-216}$$

$$= -\sqrt[3]{(2 \times 2 \times 2) \times (3 \times 3 \times 3)}$$

$$= -(2 \times 3)$$

$$= -6$$

Hence, $\sqrt[3]{-216} = -6$

Question 4(ii)

Find the cube-roots of -512.

Answer

Prime factors of 512:

2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

$\sqrt[3]{-512}$

$= -\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)}$

$= -(2 \times 2 \times 2)$

$= -8$

Hence, $\sqrt[3]{-512} = -8$

Question 4(iii)

Find the cube-roots of -1331

Answer

Prime factors of 1331:

11	1331
11	121
	11

$\sqrt[3]{-1331}$

$= -\sqrt[3]{(11 \times 11 \times 11)}$

$= -11$

Hence, $\sqrt[3]{-1331} = -11$

Question 4(iv)

Find the cube-roots of $-\dfrac{27}{125}$

Answer

Prime factors of 27:

3	27
3	9
	3

Prime factors of 125:

5	125
5	25
	5

$$\sqrt[3]{-\frac{27}{125}}$$

$$= -\frac{\sqrt[3]{27}}{\sqrt[3]{125}}$$

$$= -\frac{\sqrt[3]{(3 \times 3 \times 3)}}{\sqrt[3]{(5 \times 5 \times 5)}}$$

$$= -\frac{3}{5}$$

Hence, $\sqrt[3]{-\frac{27}{125}} = -\frac{3}{5}$

Question 4(v)

Find the cube-roots of $-\frac{64}{343}$

Answer

Prime factors of 64:

2	64
2	32
2	16
2	8
2	4
	2

Prime factors of 343:

7	343
7	49
	7

$$\sqrt[3]{-\frac{64}{343}}$$

$$= -\frac{\sqrt[3]{64}}{\sqrt[3]{343}}$$

$$= -\frac{\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)}}{\sqrt[3]{(7 \times 7 \times 7)}}$$

$$= -\frac{2 \times 2}{7}$$

$$= -\frac{4}{7}$$

Hence, $\sqrt[3]{-\frac{64}{343}} = -\frac{4}{7}$

Question 4(vi)

Find the cube-roots of $-\frac{512}{343}$

Answer

Prime factors of 512:

$$\begin{array}{c|r}
2 & 512 \\ \hline
2 & 256 \\ \hline
2 & 128 \\ \hline
2 & 64 \\ \hline
2 & 32 \\ \hline
2 & 16 \\ \hline
2 & 8 \\ \hline
2 & 4 \\ \hline
 & 2
\end{array}$$

Prime factors of 343:

$$\begin{array}{c|r}
7 & 343 \\ \hline
7 & 49 \\ \hline
 & 7
\end{array}$$

$$\sqrt[3]{-\frac{512}{343}}$$

$$= -\frac{\sqrt[3]{512}}{\sqrt[3]{343}}$$

$$= -\frac{\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)}}{\sqrt[3]{(7 \times 7 \times 7)}}$$

$$= -\frac{2 \times 2 \times 2}{7}$$

$$= -\frac{8}{7}$$

$$= -1\frac{1}{7}$$

Hence, $\sqrt[3]{-\frac{512}{343}} = -1\frac{1}{7}$

Question 4(vii)

Find the cube-roots of -2197

Answer

Prime factors of 2197:

13	2197
13	169
	13

$$\sqrt[3]{-2197}$$

$$= -\sqrt[3]{(13 \times 13 \times 13)}$$

$$= -13$$

Hence, $\sqrt[3]{-2197} = -13$

Question 4(viii)

Find the cube-roots of -5832

Answer

Prime factors of 5832:

2	5832
2	2916
2	1458
3	729
3	243
3	81
3	27
3	9
	3

$$\sqrt[3]{-5832}$$

$$= -\sqrt[3]{(2 \times 2 \times 2) \times (3 \times 3 \times 3) \times (3 \times 3 \times 3)}$$

$$= -(2 \times 3 \times 3)$$

$$= -18$$

Hence, $\sqrt[3]{-5832} = -18$

Question 4(ix)

Find the cube-roots of -2744000

Answer

Prime factors of 2744000:

2	2744000
2	1372000
2	686000
2	343000
2	171500
2	85750
5	42875
5	8575
5	1715
7	343
7	49
	7

$$\sqrt[3]{-2744000}$$

$$= -\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (5 \times 5 \times 5) \times (7 \times 7 \times 7)}$$

$$= -(2 \times 2 \times 5 \times 7)$$

$$= -140$$

Hence, $\sqrt[3]{-2744000} = -140$

Question 5(i)

Find the cube-roots of 2.744

Answer

$\sqrt[3]{2.744}$

$= \sqrt[3]{\dfrac{2744}{1000}}$

$= \dfrac{\sqrt[3]{2744}}{\sqrt[3]{1000}}$

Prime factors of 2744:

2	2744
2	1372
2	686
7	343
7	49
	7

$\therefore \dfrac{\sqrt[3]{2744}}{\sqrt[3]{1000}} = \dfrac{\sqrt[3]{(2 \times 2 \times 2) \times (7 \times 7 \times 7)}}{\sqrt[3]{10 \times 10 \times 10}}$

$= \dfrac{(2 \times 7)}{10}$

$= \dfrac{14}{10}$

$= 1.4$

Hence, $\sqrt[3]{2.744} = 1.4$

Question 5(ii)

Find the cube-roots of 9.261

Answer

$$\sqrt[3]{9.261}$$

$$= \sqrt[3]{\frac{9261}{1000}}$$

$$= \frac{\sqrt[3]{9261}}{\sqrt[3]{1000}}$$

Prime factors of 9261:

3	9261
3	3087
3	1029
7	343
7	49
	7

$$\therefore \frac{\sqrt[3]{9261}}{\sqrt[3]{1000}} = \frac{\sqrt[3]{(3 \times 3 \times 3) \times (7 \times 7 \times 7)}}{\sqrt[3]{10 \times 10 \times 10}}$$

$$= \frac{(3 \times 7)}{10}$$

$$= \frac{21}{10}$$

$$= 2.1$$

Hence, $\sqrt[3]{9.261} = 2.1$

Question 5(iii)

Find the cube-roots of 0.000027

Answer

$$\sqrt[3]{0.000027}$$

$$= \sqrt[3]{\dfrac{27}{1000000}}$$

$$= \dfrac{\sqrt[3]{27}}{\sqrt[3]{1000000}}$$

Prime factors of 27:

$$\begin{array}{r|l} 3 & 27 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

$$\therefore \dfrac{\sqrt[3]{27}}{\sqrt[3]{1000000}} = \dfrac{\sqrt[3]{(3 \times 3 \times 3)}}{\sqrt[3]{100 \times 100 \times 100}}$$

$$= \dfrac{3}{100}$$

$$= 0.03$$

Hence, $\sqrt[3]{0.000027} = 0.03$

Question 5(iv)

Find the cube-roots of -0.512

Answer

$$\sqrt[3]{-0.512}$$

$$= \sqrt[3]{-\frac{512}{1000}}$$

$$= -\frac{\sqrt[3]{512}}{\sqrt[3]{1000}}$$

Prime factors of 512:

2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

$$\therefore -\frac{\sqrt[3]{512}}{\sqrt[3]{1000}} = -\frac{\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)}}{\sqrt[3]{10 \times 10 \times 10}}$$

$$= -\frac{(2 \times 2 \times 2)}{10}$$

$$= -\frac{8}{10}$$

$$= -0.8$$

Hence, $\sqrt[3]{-0.512} = -0.8$

Question 5(v)

Find the cube-roots of -15.625

Answer

$$\sqrt[3]{-15.625}$$

$$= \sqrt[3]{-\frac{15625}{1000}}$$

$$= -\frac{\sqrt[3]{15625}}{\sqrt[3]{1000}}$$

Prime factors of 15625:

5	15625
5	3125
5	625
5	125
5	25
	5

$$= -\frac{\sqrt[3]{(5 \times 5 \times 5) \times (5 \times 5 \times 5)}}{\sqrt[3]{10 \times 10 \times 10}}$$

$$= -\frac{(5 \times 5)}{10}$$

$$= -\frac{25}{10}$$

$$= -2.5$$

Hence, $\sqrt[3]{-15.625} = -2.5$

Question 5(vi)

Find the cube-roots of -125 x 1000

Answer

$$\sqrt[3]{-125 \times 1000}$$

$$= -\sqrt[3]{125} \times \sqrt[3]{1000}$$

Prime factors of 125:

$$\begin{array}{c|c} 5 & 125 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

$$\therefore -\sqrt[3]{125} \times \sqrt[3]{1000} = -\sqrt[3]{(5 \times 5 \times 5)} \times \sqrt[3]{(10 \times 10 \times 10)}$$

$$= -5 \times 10$$

$= -50$

Hence, $\sqrt[3]{-125 \times 1000} = -50$

Question 6

Find the smallest number by which 26244 should be divided so that the quotient is a perfect cube.

Answer

Finding prime factors of 26244:

2	26244
2	13122
3	6561
3	2187
3	729
3	243
3	81
3	27
3	9
	3

$$26244 = 2 \times 2 \times (3 \times 3 \times 3) \times (3 \times 3 \times 3) \times 3 \times 3$$

Since the prime factor 2 and 3 are not in triplets.

26244 should be divided by 36 (2 x 2 x 3 x 3) so that the quotient is a perfect cube.

Question 7

What is the least number by which 30375 should be multiplied to get a perfect cube ?

Answer

Finding prime factors of 30375

3	30375
3	10125
3	3375
3	1125
3	375
5	125
5	25
	5

$$30375 = (3 \times 3 \times 3) \times 3 \times 3 \times (5 \times 5 \times 5)$$

Since the prime factor 3 is not in triplets.

3 should be multiplied with 30375 so that the product is a perfect cube.

Question 8(i)

Find the cube-roots of 700 x 2 x 49 x 5

Answer

$$\sqrt[3]{700 \times 2 \times 49 \times 5}$$

$$= \sqrt[3]{7 \times 49 \times 100 \times 2 \times 5}$$

$$= \sqrt[3]{(7 \times 7 \times 7) \times (2 \times 2 \times 2) \times (5 \times 5 \times 5)}$$

$$= 7 \times 2 \times 5$$

$$= 70$$

$$\sqrt[3]{700 \times 2 \times 49 \times 5} = 70$$

Question 8(ii)

Find the cube-roots of -216 x 1728

Answer

Prime factors of 216:

2	216
2	108
2	54
3	27
3	9
	3

Finding prime factors of 1728:

2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
	3

$\sqrt[3]{-216 \times 1728}$

$= \sqrt[3]{-216} \times \sqrt[3]{1728}$

$= -\sqrt[3]{(2 \times 2 \times 2) \times (3 \times 3 \times 3)} \times \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (3 \times 3 \times 3)}$

$= -(2 \times 3) \times (2 \times 2 \times 3)$

$= -6 \times 12$

$= -72$

$\sqrt[3]{-216 \times 1728} = -72$

Question 8(iii)

Find the cube-roots of -64 x -125

Answer

Prime factors of 64:

$$\begin{array}{c|c} 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline & 2 \end{array}$$

Prime factors of 125:

$$\begin{array}{c|c} 5 & 125 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

$$\sqrt[3]{-64 \times -125}$$

$$= \sqrt[3]{-64} \times \sqrt[3]{-125}$$

$$= -\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)} \times -\sqrt[3]{(5 \times 5 \times 5)}$$

$$= (2 \times 2) \times (5)$$

$$= 4 \times 5$$

$$= 20$$

$$\sqrt[3]{-64 \times -125} = 20$$

Question 8(iv)

Find the cube-roots of $-\dfrac{27}{343}$

Answer

Prime factors of 27:

$$\begin{array}{r|l} 3 & 27 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

Prime factors of 343:

$$\begin{array}{r|r} 7 & 343 \\ \hline 7 & 49 \\ \hline & 7 \end{array}$$

$$\sqrt[3]{-\frac{27}{343}}$$

$$= -\frac{\sqrt[3]{27}}{\sqrt[3]{343}}$$

$$= -\frac{\sqrt[3]{3 \times 3 \times 3}}{\sqrt[3]{(7 \times 7 \times 7)}}$$

$$= -\frac{3}{7}$$

Hence, $\sqrt[3]{-\dfrac{27}{343}} = -\dfrac{3}{7}$

Question 8(v)

Find the cube-roots of $\dfrac{729}{-1331}$

Answer

Prime factors of 729:

$$\begin{array}{r|r} 3 & 729 \\ \hline 3 & 243 \\ \hline 3 & 81 \\ \hline 3 & 27 \\ \hline 3 & 9 \\ \hline & 3 \end{array}$$

Prime factors of 1331:

$$\begin{array}{r|r} 11 & 1331 \\ \hline 11 & 121 \\ \hline & 11 \end{array}$$

$$\sqrt[3]{\dfrac{729}{-1331}}$$

$$= -\dfrac{\sqrt[3]{729}}{\sqrt[3]{-1331}}$$

$$= -\dfrac{\sqrt[3]{(3 \times 3 \times 3) \times (3 \times 3 \times 3)}}{\sqrt[3]{(11 \times 11 \times 11)}}$$

$$= -\frac{3 \times 3}{11}$$

$$= -\frac{9}{11}$$

Hence, $\sqrt[3]{\frac{729}{-1331}} = -\frac{9}{11}$

Question 8(vi)

Find the cube-roots of 250.047

Answer

$$\sqrt[3]{250.047}$$

$$= \sqrt[3]{\frac{250047}{1000}}$$

$$= \frac{\sqrt[3]{250047}}{\sqrt[3]{1000}}$$

Prime factors of 250047:

3	250047
3	83349
3	27783
3	9261
3	3087
3	1029
7	343
7	49
	7

$$\therefore \frac{\sqrt[3]{250047}}{\sqrt[3]{1000}} = \frac{\sqrt[3]{(3 \times 3 \times 3) \times (3 \times 3 \times 3) \times (7 \times 7 \times 7)}}{\sqrt[3]{(10 \times 10 \times 10)}}$$

$$= \frac{3 \times 3 \times 7}{10}$$

$$= \frac{63}{10}$$

$$= 6.3$$

Hence, $\sqrt[3]{250.047} = 6.3$

Question 8(vii)

Find the cube-roots of -175616

Answer

Prime factors of 175616:

2	175616
2	87808
2	43904
2	21952
2	10976
2	5488
2	2744
2	1372
2	686
7	343
7	49
	7

$\sqrt[3]{-175616}$

$= -\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (7 \times 7 \times 7)}$

$= -2 \times 2 \times 2 \times 7$

$= -56$

Hence, $\sqrt[3]{-175616} = -56$

Test Yourself

Question 1(i)

$64\sqrt{x^6}$ - $\sqrt{64 \times x^6}$ is equal to :

1. $72x^3$
2. $56x^2$
3. $128x^3$
4. $56x^3$

Answer

$$64\sqrt{x^6} - \sqrt{64 \times x^6}$$

$$= 64 \times x^3 - \sqrt{64} \times \sqrt{x^6}$$

$$= 64 \times x^3 - 8 \times x^3$$

$$= (64 - 8) \times x^3$$

$= 56 \times x^3$

Hence, option 4 is the correct option.

Question 1(ii)

If a number is multiplied by 3, its square will be multiplied by :

1. 9
2. 3
3. 27
4. 81

Answer

Lets the number be x.

Number multiplied by 3 = $3x$.

Square of $3x$ = $(3x)^2$

$= (3 \times x)^2$

$= 3^2 \times x^2$

$= 9 \times x^2$

Hence, option 1 is the correct option.

Question 1(iii)

Two numbers are in the ratio 5 : 4. If the difference of their cubes is 61; the numbers are :

1. 5 and 4
2. 25 and 16
3. 10 and 18
4. none of these

Answer

Ratio of two numbers are 5:4.

Lets the numbers be $5x$ and $4x$. Hence,

$(5x)^3 - (4x)^3 = 61$

$\Rightarrow 125x^3 - 64x^3 = 61$

$\Rightarrow (125 - 64)x^3 = 61$

$\Rightarrow 61x^3 = 61$

$\Rightarrow x^3 = \dfrac{61}{61}$

$\Rightarrow x^3 = 1$

$\Rightarrow x = \sqrt[3]{1}$

$\Rightarrow x = 1$ So the numbers are 5 and 4.

Hence, option 1 is the correct option.

Question 1(iv)

The value of $\sqrt[3]{27} + \sqrt[3]{0.008} + \sqrt[3]{0.064}$ is :

1. $\sqrt[3]{27.072}$
2. 3.72
3. 3.6
4. 3.06

Answer

$$\sqrt[3]{27} + \sqrt[3]{0.008} + \sqrt[3]{0.064}$$

$$= \sqrt[3]{(3 \times 3 \times 3)} + \sqrt[3]{\frac{8}{1000}} + \sqrt[3]{\frac{64}{1000}}$$

$$= 3 + \frac{2}{10} + \frac{4}{10}$$

$$= 3 + 0.2 + 0.4$$

$$= 3.6$$

Hence, option 3 is the correct option.

Question 1(v)

The value of $\sqrt[3]{(-3)^3 \times 8}$ is :

1. -27

2. -6
3. 6
4. none of these

Answer

$\sqrt[3]{(-3)^3 \times 8}$

$= \sqrt[3]{(-3)^3} \times \sqrt[3]{8}$

$= -3 \times 2$

$= -6$

Hence, option 2 is the correct option.

Question 1(vi)

Statement 1: Cubes of all odd natural numbers are odd.

Statement 2: Cubes of negative integers are positive or negative integers.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

An odd number can be represented as 2n + 1, where n is an integer.

Cube of odd number = $(2n + 1)^3$

$\Rightarrow (2n)^3 + 1^3 + 3 \times 2n \times 1 \times (2n + 1)$

$\Rightarrow 8n^3 + 1 + 6n(2n + 1)$

$\Rightarrow 8n^3 + 1 + 12n^2 + 6n$

$\Rightarrow 8n^3 + 12n^2 + 6n + 1$ ....(1)

If any no. odd or even is multiplied by an even number it becomes an even number.

Since, 8, 12 and 6 are even numbers so first three terms of equation (1) are even, and adding 1 at the end ensures the result is odd.

So, statement 1 is true.

Let's take some negative number, -2 and -3.

Cube of -2 = $(-2)^3 = -8$

Cube of -3 = $(-3)^3 = -27$

The cube of a negative integer is always a negative integer.

So, statement 2 is false.

Hence, Option 3 is the correct option.

Question 1(vii)

Assertion (A) : The smallest number by which 1323 may be multiplied so that the product is a perfect cube of 7.

Reason (R) : A given natural number is a perfect cube if in its prime factorization every prime occurs three times.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Prime Factorization of 1323 = (3 x 3 x 3) x 7 x 7 = 3^3 x 7^2

On multiplying by 1323 by 7, we get :

⇒ 1323 x 7 = (3 x 3 x 3) x 7 x 7 x 7

⇒ 1323 x 7 = 3^3 x 7^3

⇒ 1323 x 7 = $(3 \text{ x } 7)^3$

⇒ 1323 x 7 = 21^3

∴ On multiplying 1323 by 7, it becomes a perfect cube.

So, assertion (A) is true.

In the prime factorization of a perfect cube, each prime must appear with an exponent that is a multiple of 3, it is not necessary that every prime number occurs only three times.

So, reason (R) is false.

Hence, option 3 is the correct option.

Question 1(viii)

Assertion (A) : $\sqrt[3]{4\dfrac{12}{125}} = 1\dfrac{3}{5}$

Reason (R) : If p and q are two whole numbers (p ≠ 0), then $\sqrt[3]{\dfrac{p}{q}} = \dfrac{\sqrt[3]{p}}{\sqrt[3]{q}}$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

If p and q are two whole numbers (p ≠ 0), then $\sqrt[3]{\dfrac{p}{q}} = \dfrac{\sqrt[3]{p}}{\sqrt[3]{q}}$.

This is a fundamental property of radicals.

So, reason (R) is true.

Solving,

$$\Rightarrow \sqrt[3]{4\frac{12}{125}}$$

$$\Rightarrow \sqrt[3]{\frac{4 \times 125 + 12}{125}}$$

$$\Rightarrow \sqrt[3]{\frac{500 + 12}{125}}$$

$$\Rightarrow \sqrt[3]{\frac{512}{125}}$$

$$\Rightarrow \frac{\sqrt[3]{512}}{\sqrt[3]{125}}$$

$$\Rightarrow \frac{8}{5}$$

$$\Rightarrow 1\frac{3}{5}$$

So, assertion (A) is true and reason (R) clearly explains assertion.

Hence, option 1 is the correct option.

Question 1(ix)

Assertion (A) : $\sqrt[3]{-125} = \pm 25$

Reason (R) : The cube root of a negative perfect cube is negative.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

⇒ -125 = -5 x -5 x -5

⇒ -125 = $(-5)^3$

⇒ $\sqrt[3]{-125} = -5.$

So, assertion (A) is false.

We know that,

The cube root of a negative perfect cube is always a negative number.

So, reason (R) is true.

Hence, option 4 is the correct option.

Question 1(x)

Assertion (A) : $\sqrt[3]{968} \times \sqrt[3]{1375} = 110$

Reason (R) : If p and q are two whole numbers, then $\sqrt[3]{p} \times \sqrt[3]{q} = \sqrt[3]{pq}$.

1. Both A and R are correct, and R is the correct explanation for A.

2. Both A and R are correct, and R is not the correct explanation for A.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

If p and q are two whole numbers, then $\sqrt[3]{p} \times \sqrt[3]{q} = \sqrt[3]{p \times q} = \sqrt[3]{pq}$.

So, reason (R) is true.

Solving,

$\Rightarrow \sqrt[3]{968} \times \sqrt[3]{1375}$

$\Rightarrow \sqrt[3]{968 \times 1375}$

$\Rightarrow \sqrt[3]{1331000}$

$\Rightarrow 110.$

So, assertion (A) is true and reason (R) clearly explains assertion.

Hence, option 1 is the correct option.

Question 2

State true or false :

(i) Cube of an odd number can be even.

(ii) A perfect cube does not end with two zeroes.

(iii) If square of a number ends with 5, its cube will end with 25.

(iv) The cube of a two digit number may be a three digit number.

(v) Cube of a natural number is called perfect cube.

Answer

(i) False.

Reason — Cube of an odd number are odd.

(ii) True.

Reason — A perfect cube always end in the number of zeroes that is a multiple of 3.

(iii) False.

Reason — It is not necessary that if square of a number ends with 5, its cube will end with 25.
Example- The square of 15 is 225 but its cube is 3375. The square of 5 is 25 and its cube is 125.

(iv) False.

Reason — Cube of two digit number may have four digits to six digits.

(v) True.

Reason — The cube of a number is called a perfect cube.

Question 3(i)

Find the cube root of 110.592

Answer

$$\sqrt[3]{110.592}$$

$$= \sqrt[3]{\frac{110592}{1000}}$$

$$= \frac{\sqrt[3]{110592}}{\sqrt[3]{1000}}$$

Prime factors of 110592:

2	110592
2	55296
2	27648
2	13824
2	6912
2	3456
2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
	3

$$\therefore \frac{\sqrt[3]{110592}}{\sqrt[3]{1000}} = \frac{\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (3 \times 3 \times 3)}}{\sqrt[3]{(10 \times 10 \times 10)}}$$

$$= \frac{2 \times 2 \times 2 \times 2 \times 3}{10}$$

$$= \frac{48}{10}$$

$$= 4.8$$

Hence, $\sqrt[3]{110.592} = 4.8$

Question 3(ii)

Find the cube root of 0.064

Answer

$\sqrt[3]{0.064}$

$= \sqrt[3]{\dfrac{64}{1000}}$

$= \dfrac{\sqrt[3]{64}}{\sqrt[3]{1000}}$

Prime factors of 64:

2	64
2	32
2	16
2	8
2	4
	2

$\therefore \dfrac{\sqrt[3]{64}}{\sqrt[3]{1000}} = \dfrac{\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)}}{\sqrt[3]{(10 \times 10 \times 10)}}$

$= \dfrac{2 \times 2}{10}$

$$= \frac{4}{10}$$

$$= 0.4$$

Hence, $\sqrt[3]{0.064} = 0.4$

Question 4

Find the volume of a cubical box whose surface area is 486 cm^2.

Answer

Let a be the length of cubical box.

Surface area of cubical box = 6a^2

$$6a^2 = 486$$

$$\Rightarrow a^2 = \frac{486}{6}$$

$$\Rightarrow a^2 = 81$$

$$\Rightarrow a = \sqrt{81}$$

$$\Rightarrow a = 9$$

Volume of cubical box = a^3

$$(9)^3$$

$$= 9 \times 9 \times 9$$

$= 729$

The volume of cubical box is 729 cm^3.

Question 5(i)

Find cube root of 125 x -64

Answer

Prime factors of 125:

$$\begin{array}{r|l} 5 & 125 \\ \hline 5 & 25 \\ \hline & 5 \end{array}$$

Prime factors of 64:

$$\begin{array}{r|l} 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline & 2 \end{array}$$

$\sqrt[3]{125 \times -64}$

$= \sqrt[3]{125} \times \sqrt[3]{-64}$

$= \sqrt[3]{(5 \times 5 \times 5)} \times -\sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2)}$

$= (5) \times -(2 \times 2)$

$= 5 \times -4$

$= -20$

Hence, $\sqrt[3]{125 \times -64} = -20$

Question 5(ii)

Find cube root of $\dfrac{-125}{343}$

Answer

Prime factors of 125:

5	125
5	25
	5

Prime factors of 343:

7	343
7	49
	7

$\sqrt[3]{\dfrac{-125}{343}}$

$= \dfrac{\sqrt[3]{-125}}{\sqrt[3]{343}}$

$$= -\frac{\sqrt[3]{(5 \times 5 \times 5)}}{\sqrt[3]{(7 \times 7 \times 7)}}$$

$$= -\frac{5}{7}$$

Hence, $\sqrt[3]{\frac{-125}{343}} = -\frac{5}{7}$

Question 6

Three numbers are in the ratio 2 : 3 : 1. The sum of their cubes is 288. Find the numbers.

Answer

Three numbers are in the ratio 2 : 3 : 1. So the three numbers are $2x, 3x$ and $1x$. Hence,

$$(2x)^3 + (3x)^3 + (1x)^3 = 288$$

$$\Rightarrow 8x^3 + 27x^3 + 1x^3 = 288$$

$$\Rightarrow 36x^3 = 288$$

$$\Rightarrow x^3 = \frac{288}{36}$$

$$\Rightarrow x^3 = 8$$

$$\Rightarrow x = \sqrt[3]{8}$$

$$\Rightarrow x = \sqrt[3]{2 \times 2 \times 2}$$

$\Rightarrow x = 2$

2x = 2 x 2 = 4

3x = 3 x 2 = 6

Hence, the three numbers are 4, 6 and 2

Question 7

Find the smallest number by which 14,580 must be multiplied to make a perfect cube. Also, find the cube root of the perfect cube number obtained.

Answer

Finding prime factors of 14580

$$\begin{array}{r|r} 2 & 14580 \\ \hline 2 & 7290 \\ \hline 3 & 3645 \\ \hline 3 & 1215 \\ \hline 3 & 405 \\ \hline 3 & 135 \\ \hline 3 & 45 \\ \hline 3 & 15 \\ \hline & 5 \end{array}$$

$$14580 = 2 \times 2 \times (3 \times 3 \times 3) \times (3 \times 3 \times 3) \times 5$$

Since the prime factor 2 and 5 are not in triplets,

Hence, 14,580 must be multiplied with 2 x 5 x 5 = 50

$$14580 \times 50 = 729000$$

$$\sqrt[3]{729000} = \sqrt[3]{(2 \times 2 \times 2) \times (3 \times 3 \times 3) \times (3 \times 3 \times 3) \times (5 \times 5 \times 5)}$$

$$\sqrt[3]{729000} = \sqrt[3]{2 \times 3 \times 3 \times 50}$$

$$\sqrt[3]{729000} = 90$$

14,580 should be multiplied with 50 so that the product is a perfect cube. The cube root of 729000 is 90.

Question 8

Find the smallest number by which 8,232 must be divided to make it a perfect cube. Also, find the cube root of the perfect cube so obtained.

Answer

Finding prime factors of 8232

2	8232
2	4116
2	2058
3	1029
7	343
7	49
	7

$$8232 = (2 \times 2 \times 2) \times 3 \times (7 \times 7 \times 7)$$

Since the prime factor 3 is not in triplet, so 8,232 must be divided by 3 to make it a perfect cube.

$$\frac{8232}{3} = 2744$$

Finding prime factors of 2744

2	2744
2	1372
2	686
7	343
7	49
	7

$$\sqrt[3]{2744} = \sqrt[3]{(2 \times 2 \times 2) \times (7 \times 7 \times 7)}$$

$$\sqrt[3]{2744} = \sqrt[3]{2 \times 7}$$

$$\sqrt[3]{2744} = 14$$

2744 must be divided by 3 so that the quotient is a perfect cube. The cube root of 2744 is 14.

Question 9(i)

Evaluate $\left[(12^2 + 5^2)^{\frac{1}{2}}\right]^3$

Answer

$$\left[(12^2 + 5^2)^{\frac{1}{2}}\right]^3$$

$$= \left[(144 + 25)^{\frac{1}{2}}\right]^3$$

$$= \left[(169)^{\frac{1}{2}}\right]^3$$

$$= \left[(13^2)^{\frac{1}{2}}\right]^3$$

$$= \left[(13)^{\frac{2}{2}}\right]^3$$

$$= [(13)]^3$$

$$= 13 \times 13 \times 13$$

$$= 2197$$

Hence, $\left[(12^2 + 5^2)^{\frac{1}{2}}\right]^3 = 2197$

Question 9(ii)

Evaluate $(\sqrt{10^3 - 6^3})^3$

Answer

$(\sqrt{10^3 - 6^3})^3$

$= (\sqrt{1000 - 216})^3$

$= (\sqrt{784})^3$

Finding prime factors of 784

2	784
2	392
2	196
2	98
7	49
	7

$\therefore (\sqrt{784})^3 = (\sqrt{(2 \times 2) \times (2 \times 2) \times (7 \times 7)})^3$

$= (2 \times 2 \times 7)^3$

$= (28)^3$

$= 21952$

Hence, $(\sqrt{10^3 - 6^3})^3 = 21952$

Question 10

Difference of two perfect cubes is 387. If the cube root of the greater of the two numbers is 8, find the cube root of the smaller number.

Answer

Cube root of greater number = 8.

Let the cube root of smaller number be x.

Hence,

$8^3 - x^3 = 387$

$\Rightarrow 512 - x^3 = 387$

$\Rightarrow 512 - 387 = x^3$

$\Rightarrow 125 = x^3$

$\Rightarrow x = \sqrt[3]{125}$

$\Rightarrow x = \sqrt[3]{5 \times 5 \times 5}$

$\Rightarrow x = 5$

Hence, the cube root of the smaller number is 5

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